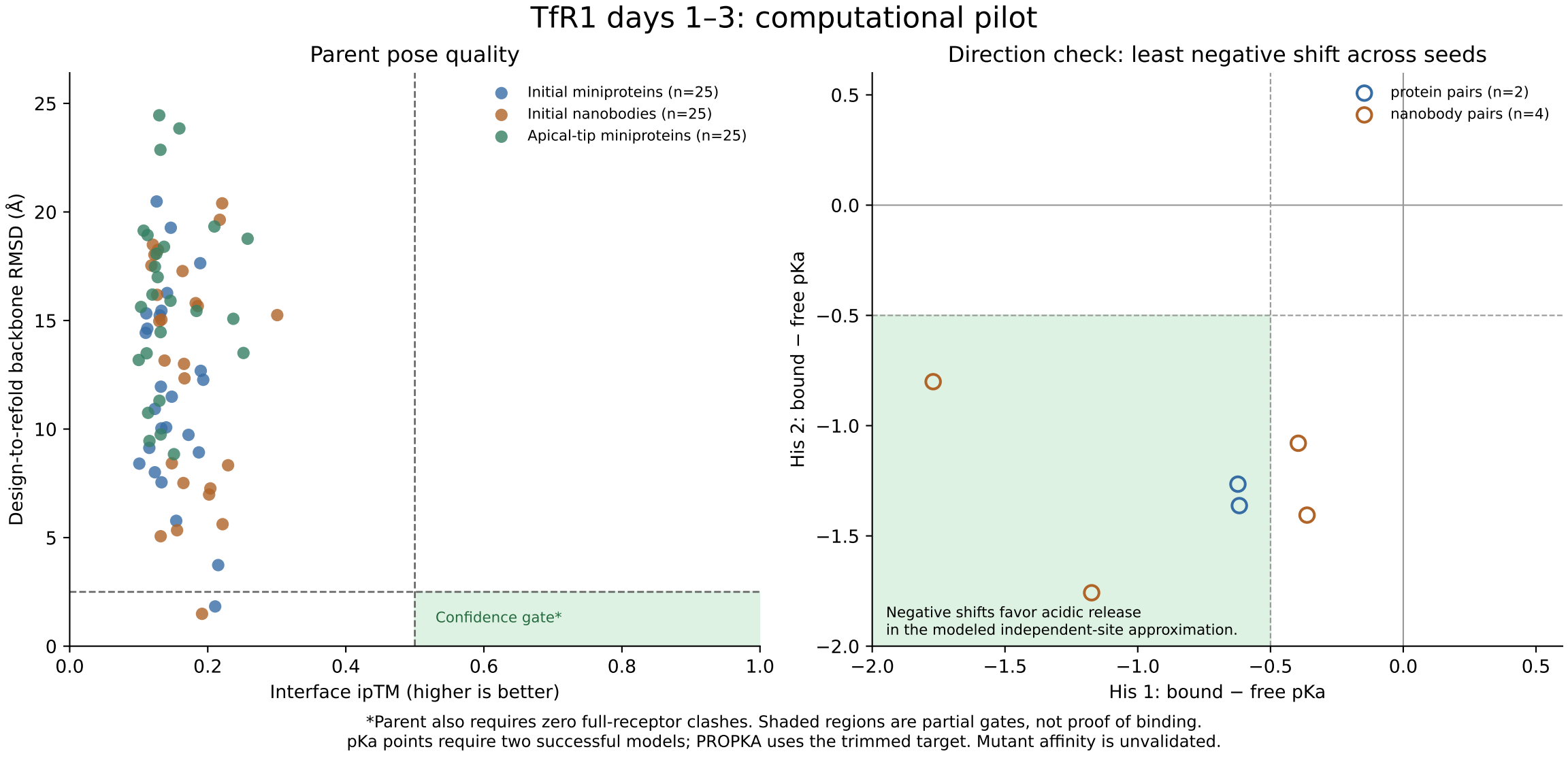
TfR1 days 1–3: computational pilot
Parent pose quality
Initial miniproteins (n=25)
Initial nanobodies (n=25)
Apical-tip miniproteins (n=25)
Design-to-refold backbone RMSD (Å)
Interface ipTM (higher is better)
Confidence gate*
Direction check: least negative shift across seeds
protein pairs (n=2)
nanobody pairs (n=4)
His 2: bound − free pKa
His 1: bound − free pKa
Negative shifts favor acidic release
in the modeled independent-site approximation.
*Parent also requires zero full-receptor clashes. Shaded regions are partial gates, not proof of binding.
pKa points require two successful models; PROPKA uses the trimmed target. Mutant affinity is unvalidated.